

DOUGLAS GUILHERME

BIOINFORMATICS MACHINE LEARNING SPECIALIST

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PROFESSIONAL SUMMARY

Bioinformatics Machine Learning Specialist with 5+ years of production-grade experience building and—crucially—evaluating AI-driven analytical and predictive models for precision medicine and biopharmaceutical discovery. I bridge the gap between computational genomics and clinical validation, with deep expertise in identifying failure modes of AI/LLMs in biological reasoning—including variant prioritization, transcriptomic pathway inference, and protein interaction prediction.

Owns bioinformatics pipelines end-to-end: from raw sequencing data processing and feature engineering to model deployment, clinical interpretation, and rigorous human-in-the-loop auditing. Partners with research scientists, clinicians, and cross-functional stakeholders to validate AI outputs against gold-standard biological benchmarks. Ships scalable analytical platforms quickly, iterates from biological validation metrics, and maintains high-availability genomic data infrastructure with Docker, cloud computing, and robust observability tooling.

Core focus: Auditing and annotating AI-generated genomic interpretations to catch false-positive pathogenic calls, correct hallucinated pathway inferences, and provide structured human reasoning for model retraining.

CORE COMPETENCIES

AI/LLM Evaluation & Auditing: Stress-testing model reasoning in genomic diagnostics, variant pathogenicity validation, causal inference auditing, structured human reasoning annotation, hallucination detection in biological contexts

Bioinformatics: RNA-Seq Analysis, Variant Calling, Genome Assembly, Metagenomics, Transcriptomics, Proteogenomics, Molecular Docking, Virtual Screening, Microbiome Profiling

Machine Learning & AI: Deep Learning, Neural Networks, Clustering (k-means, DBSCAN, Mean Shift), Predictive Modeling, Feature Engineering, Hyperparameter Optimization, Dimensionality Reduction, Statistical Analysis

Programming & Workflows: Python, R, SQL, Bash/Shell, JavaScript, Nextflow, Bioconductor, QIIME2

ML Frameworks: TensorFlow, PyTorch, Scikit-learn, Jupyter

Cloud & Infrastructure: Google Cloud Platform (GCP), AWS, Docker, Kubernetes, Git, Prometheus, Grafana, BigQuery

Domains: Precision Medicine, Oncology Diagnostics, Drug Discovery, Infectious Disease Genomics, Public Health Genomics, Regulatory Compliance (LGPD, HIPAA)

WORK EXPERIENCE

Bioinformatics Machine Learning Specialist

Jan 2024 – Present (2026)

Nita Alimentos official – Florianópolis, Brazil

- Led end-to-end delivery of enterprise-scale machine learning pipelines for digital microbiological diagnostics, automating high-throughput sequencing data ingestion and aligning business requirements across R&D, Quality Assurance, and international stakeholders over 3 major production releases.
- Architected and deployed deep neural networks for real-time contamination risk prediction in industrial production environments, engineering versioned model registries and validation frameworks that reduced false-positive flags by 32% and saved 150+ hours of manual review monthly.
- Audited AI-generated variant classification and contamination risk reports against validated biological benchmarks (NCBI, ClinVar, curated pathway libraries), identifying false-positive inferences in 15% of automated runs and establishing structured human reasoning loops for continuous model retraining.

- Designed robust evaluation protocols to benchmark LLM reasoning in complex genomic interpretations, pinpointing hallucination patterns in low-coverage sequencing data and applying expert corrective annotations to improve precision.
- Engineered responsive, analytical UI dashboards from wireframes, incorporating intuitive data visualizations and UX refinements that reduced diagnostic decision time by 40% for quality control teams.
- Architected relational PostgreSQL schemas and optimized complex SQL migration scripts for high-dimensional genomic metadata, achieving sub-second query performance across a database of 500,000+ sequencing samples.
- Orchestrated microservice containerization using Docker and Kubernetes, building fault-tolerant CI/CD deployment pipelines that achieved 99.9% operational uptime across production and staging environments.
- Implemented scalable, event-driven data workflows using Nextflow, driving a 40% reduction in end-to-end processing latency compared to legacy batch architectures.
- Configured comprehensive Prometheus and Grafana monitoring stacks with SLI/SLA alerts for pipeline latency and node failure rates, resolving 85% of system regressions prior to user impact.
- Mentored 2 junior bioinformaticians on clean code principles, rigor in unit/integration testing, and best practices for auditing AI outputs, elevating automated test coverage from 55% to 78%.

AI/ML Research Associate – Transcriptomics

Oct 2023 – Dec 2023

Mosten – Sao Paulo, Brazil

- Architected full-stack analytical features for the flagship BDASeq transcriptomics platform, owning predictive ML models and cloud infrastructure from conceptual prototype to high-availability production rollout.
- Engineered cloud-native ingestion pathways on Google Cloud Platform (GCP) for parallelized RNA-Seq processing, integrating automated error handling and idempotent logic that completely eliminated duplicate job executions.
- Optimized BigQuery analytical queries, materialization, and database indexing, slashing differential expression report execution times from 12 seconds down to 2.3 seconds for large-scale transcriptomic datasets.
- Partnered closely with clinical collaborators and research scientists in a high-velocity async environment to define MVP milestones, delivering biweekly production iterations with zero critical post-release defects.
- Authored developer-facing API documentation, interactive Jupyter notebook guides, and integration specs, accelerating partner onboarding and cutting time-to-first-analysis by 40%.
- Introduced automated testing frameworks and CI checks across core biological modules, decreasing production regression defects by 45% over a 4-month span.

Genomic Data Scientist

Jan 2022 – Sep 2023

DNA GTx Bioinformatics – Natal, Brazil

- Engineered high-performance React and Python components for the D-Krypt variant prioritization engine, accelerating clinical variant triage by 35% via intelligent caching, vectorization, and memoized computational workflows.
- Translated complex genomic criteria into intuitive diagnostic workflows, attaining a 95% clinician satisfaction rating on primary variant filtering interfaces.
- Co-designed RESTful API contract specifications with core backend teams for high-throughput variant annotation endpoints, halving frontend integration friction and technical debt.
- Pioneered a modular, domain-driven UI component library adopted across 3 engineering squads, standardizing visualization patterns for genomic variant calling and annotation.
- Actively contributed to agile sprint ceremonies and client demos, successfully shipping 20+ production-grade analytical features with a 100% on-time milestone delivery track record over 20 months.

Machine Learning Engineer – Structural Genomics

Aug 2020 – Dec 2021

Structural Genomics Laboratory (GenEstrut) – UFPel – Pelotas, Brazil

- Developed ML models for structural and functional genomics analysis, processing genomic and transcriptomic data for precision medicine and environmental genomics research applications.
- Built bioinformatics pipelines for microRNA biology and epigenetic data analysis, supporting patent development for molecular-based applications licensed to biotechnology companies.

- Contributed to Molecular Diagnostic Unit during COVID-19 pandemic, processing 10,000+ molecular tests and supporting regional public health response.
- Collaborated with 5+ graduate researchers to implement scalable genomic data processing workflows, reducing analysis bottlenecks and accelerating research outcomes.

EDUCATION

Master of Science in Bioinformatics

Universidade Federal do Rio Grande do Norte (UFRN) – Natal, Brazil

All core coursework completed; research focus: Integration of AI and transcriptomic data for molecular target identification (thesis pending)

Curso Superior de Tecnologia (CST), Análise e Desenvolvimento de Sistemas

Fatec Baixada Santista Rubens Lara – Diploma Tecnólogo | Feb 2022 – Jun 2025

Desenvolvimento & Data: Full-stack (Python, C#, Java, JavaScript, Angular), modelagem SQL/NoSQL e pipelines ETL.

IA & Analytics: Fundamentos de Machine Learning/IA (Scikit-learn, TensorFlow) e Big Data (Spark).

Infraestrutura: Práticas de cibersegurança, segurança de redes e programação lógica.

Bachelor's Degree in Biotechnology

Universidade Federal de Pelotas (UFPEl) – Pelotas, Brazil | 2017–2021

Graduated with honors. Emphasis on structural genomics and computational biology.

PUBLICATIONS

Enhanced Transcriptomic Discovery of Druggable Targets for Huntington's Disease – Brain Sciences, 2025
(Contributing Author)

BDASeq: A Scalable Analytical-Predictive Model for RNA-Seq Data Analysis – Presented at MENA Rare Disease Congress, UAE, 2024

Machine Learning Approaches in Precision Oncology Diagnostics – Manuscript under review, Journal of Computational Biology

CERTIFICATIONS

Machine Learning Specialization – Stanford University / Coursera

Deep Learning Specialization – deeplearning.ai

Genomic Data Science Specialization – Johns Hopkins University / Coursera

Nextflow & Scalable Bioinformatics Workflows – Seqera Labs